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COURSE :

Artificial Intelligence

CODE :

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1)

Map Colouring CSP

Regions A, B, C and D form a cycle. A is adjacent to B and C; B to A and D; C to A and D; D to B and C.

Use Red, Green and Blue and solve using backtracking.

Region	Available colours	Assigned colour
A	R G B	Red
B	R G B	Green
C	R G B	Green
D	R G B	Red

2.

Question 16 - Cryptarithmic CSP

$SEND + MORE = MONEY$. Each letter represents a unique digit from 0-9 and M cannot be 0, perform constraint propagation, find a valid assignment and verify the addition

Answer:

$$\begin{array}{r} SEND \\ + MORE \\ \hline MONEY \end{array}$$

$$S = 9$$

$$E = 5$$

$$N = 6$$

$$D = 7$$

$$M = 1$$

$$O = 0$$

$$R = 8$$

$$Y = 2$$

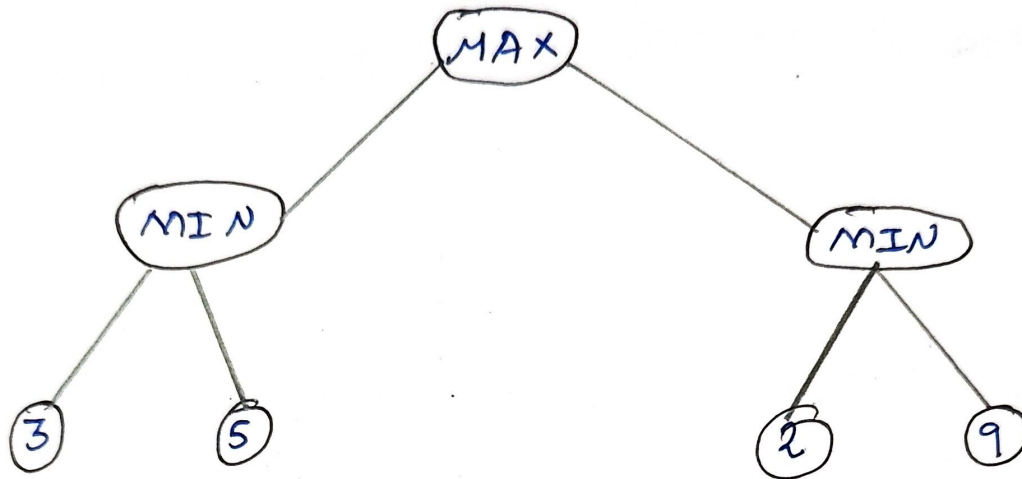
Therefore:

$$\begin{array}{r} 9567 \\ 1085 \\ + \hline 10652 \end{array}$$

3) Minimax Dry Run

(2)

perform the Minimax algorithm on the displayed game tree,
 find,
 calculate the value of each MIN node, the MAX node value and the
 move selected by MAX.



Node	children values	value
1.		
2		
3		
4		
5		